ACHYUT ANANT DESHPANDE

Mechanical Engineer | Mechanical Design | Material Handling | Manufacturing Drawings

Pune, Maharashtra | 7420878362 | deshpandeachyut27@gmail.com | LinkedIn: www.linkedin.com/in/achyut-deshpande-b34369334 | Portfolio: achyutdeshpande.github.io

PROFESSIONAL SUMMARY

Mechanical Engineer with hands-on experience in mechanical design, CAD modelling, engineering drawings, and bulk material handling equipment. Experienced in 2D/3D modelling, manufacturing & GA drawings, detailed drawings, and conveyor design calculations for screw conveyors and feeders. Skilled in SolidWorks, AutoCAD, Fusion 360, and Autodesk Inventor.

PROFESSIONAL EXPERIENCE

Mechanical Design Intern | *Screwtech Engineering Pvt. Ltd.* | *Jun 2026 – October 2026*

- Create 2D and 3D CAD models for bulk material-handling equipment including Screw Feeders, Screw Conveyors, & Mechanical Pusher Feeders.
- Prepare manufacturing, fabrication and detailed drawings with dimensions and design information required for component production and assembly.
- Perform engineering calculations and determine design values for conveyor systems based on project requirements.
- Develop General Arrangement (GA) drawings for complete conveyor and feeder assemblies.
- Prepare detailed component and assembly drawings to communicate design intent and support manufacturing activities.
- Modify and update CAD models and drawings according to project requirements and design changes.

Quality Control Intern | *Rotadyne Precision Engineering Pvt. Ltd.* | *Dec 2024 – Dec 2024*

- Gained practical exposure to quality inspection, dimensional measurement, and quality control processes in a precision engineering environment.

PROJECTS

Thermochemical Energy Storage (TCES) System — Final-Year Project:

- Investigated thermochemical energy storage using salt hydrates, including $SrBr_2$ and $SrCl_2$, with focus on energy storage behavior and thermal performance.
- Designed and tested a reactor concept to study charging/discharging behavior, hydration/dehydration response and thermal efficiency.
- Conducted experimental testing and analyzed temperature response during dehydration and hydration cycles.
- Awarded the Best Project Award (2025–26) by DKTE Society's Textile & Engineering Institute.

Air-Based Rain Protection Helmet:

- Developed a conceptual helmet with an air-inflated dome for rain protection, focusing on structural feasibility, airflow stability and ergonomics; design patent filed, Application No. 454613-001.

Water Surface Waste Collector:

- Designed a stationary floating waste-collection concept for ponds and small water bodies, emphasizing low-cost materials, passive operation and floating-debris removal.

TECHNICAL SKILLS

CAD & Drafting: SolidWorks, AutoCAD 2D/3D, Fusion 360, Autodesk Inventor.

Mechanical Design: 2D/3D CAD Modelling, General Arrangement (GA) Drawings, Manufacturing Drawings, Detailed Drawings, Conveyor Design Calculations.

Other Tools: Microsoft Office, Python (Basic), C (Basic).

EDUCATION

B.Tech. Mechanical Engineering: DKTE Society's Textile & Engineering Institute. | 2022 – 2026

CERTIFICATIONS

- CSWA: Certified SOLIDWORKS CAD Design Associate.
- Tata Technologies Ready Engineer: Advanced Automobile Design, Analysis & Manufacturing.
- Six Sigma White Belt: Knowledge of continuous improvement, waste reduction, and basic quality tools.
- NPTEL Certification: Advanced Machining Processes.